Probability and Statistics Midterm Solutions

1. In a pool of 27 applicants for a job, 20 can track, and 15 can do accounting. How many can both track and do accounting?

Solution

$20 + 15 - 27 = 8$, so 8.

2. Let C_1, C_2, C_3 be independent events with probabilities $1/2, 1/5$ and $1/5$ respectively. What is $P(C_1 \cup C_2 \cup C_3)$?

Solution

The probability that none of the events happens is $1/2 * 4/5 * 4/5 = 8/25$. So the probability that at least one did is $17/25$.

3. Let X_1 and X_2 have joint pdf $h(x_1, x_2) = cx_1^2 x_2$ where $0 < x_1 < x_2 < 1$ and 0 elsewhere. Find c, and then find the joint pdf of $Y_1 = X_1/X_2$ and $Y_2 = X_2$.

Solution

We get $c = 15$ by solving

$$\begin{aligned} 1 &= \int_0^1 \int_{x_1}^1 cx_1^2 x_2 dx_2 dx_1 = c \int_0^1 [\frac{1}{2} x_1^2 x_2^2]_{x_1}^1 dx_1 \\ &= c/2 \int_0^1 x_1^2 - x_1^4 dx_1 = \frac{c}{2}[\frac{1}{3}x_1^3 - \frac{1}{5}x_1^5]_0^1 = \frac{c}{2}\frac{2}{15} = \frac{c}{15} \end{aligned}$$

The transformation $Y_1 = u_1(X_1, X_2)$ and $Y_2 = u_2(X_1, x_2)$ has inverse transformation $X_1 = w_1(Y_1, Y_2) = Y_1 Y_2$ and $X_2 = w_2(Y_1, Y_2) = Y_2$. Thus the joint pdf f of Y_1 and Y_2 is

$$f(y_1, y_2) = h(w_1(y_1, y_2), w_2(y_1, y_2)) \cdot |J| = 15 y_1^2 y_2^3 \cdot |J|.$$

Where the jacobian J is

$$J = (\delta w_1/\delta y_1)(\delta w_2/\delta y_2) - (\delta w_1/\delta y_2)(\delta w_2/\delta y_1) = y_2 - y_1(0) = y_2.$$

Thus $f(y_1, y_2) = 15 y_1^2 y_2^4$.

4. Let X and Y have joint pdf $f(x, y) = 4xy$ on the support $0 < x, y < 1$.

(a) Find $E(XY)$.

(b) Find the marginal pdfs f_X and f_Y.

(c) Find the conditional pdf $f_{X|Y}(x, y)$.

(d) Describe the random variable $E(X|Y)$.

Solution

(a) $E(XY) = \int_0^1 \int_0^1 xy \cdot 4xy \cdot dydx = 4/9$.

(b) $f_X(x) = \int_0^1 4xy \cdot dy = 2x$ and similarily $f_Y(y) = 2y$.

(c) $f_{X|Y}(x, y) = f(x, y)/f_Y(y) = 4xy/2y = 2x$.

(d) For any $y \in (0, 1)$ the expected value $E(X|y)$ is

$$E(X|y) = \int_0^1 x \cdot f_{X|Y}(x|y) \cdot dx = \int_0^1 2x^2 dx = 2/3$$

so $E(X|Y)$ is a discrete random variable that takes the value $2/3$ with probability 1.

5. Let X and Y have joint pdf $f(x, y) = 2$ when $0 < x < y < 1$ and $f(x, y) = 0$ elsewhere. Find the mean and varience of the conditional distribution of Y given that $X = x$.

Solution

The conditional pdf $f = f_{Y|X}$ for $X = x$ is clearly $f(y) = 1/(1 - x)$ as it is a uniform distribution over a segment of length $1 - x$. We can also see this by finding the marginal pdf f_X of X by integrating over y

$$f_X(x) = \int_x^1 2dy = 2(1 - x)$$

and then using the defintion of $f_{Y|X}$:

$$f_{Y|X}(y, x) = \frac{f_{Y,X}(y, x)}{f_X(x)} = \frac{2}{2(1 - x)} = \frac{1}{1 - x}.$$

The mean of this uniform distribution is clearly the middle of its support: $x + 1/2(1 - x) = (1 + x)/2$. Or again we show this rigourously.

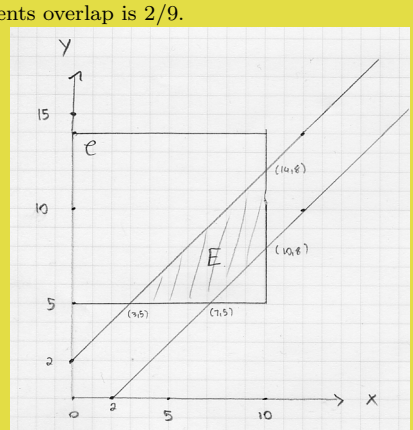
$$\mu = \int_x^1 \frac{y}{1 - x} dy = \frac{1}{1 - x}[\frac{y^2}{2}]_x^1 = \frac{1 - x^2}{2(1 - x)} = \frac{1 + x}{2}.$$

To compute the varience, we compute the second moment:

$$E((Y|x)^2) = \int_x^1 \frac{y^2}{1 - x} dy = \frac{1 - x^3}{3(1 - x)}$$

So $\sigma^2 = \frac{1-x^3}{3(1-x)} - (\frac{1+x}{2})^2$.

6. Two line segments, each two units long, are placed along the x-axis. The midpoint of the first is between $x = 0$ and $x = 10$ with uniform distribution and that of the second is between $x = 5$ and $x = 14$ uniform distribution. Assuming that the distributions of the midpoints are independent, what it the probability that the line segments overlap.

Solution

Let X be the random variable describing the midpoint of the first segment and Y that of the second. Then (X, Y) is uniform with sample space $C = [0, 10] \times [5, 14]$ having area 90. The event E that the segments overlap is the intersection of the sample space with the region $\{(x, y) \mid y \in [x - 2, x + 2]\}$. A little grade school geometry yields that E has area 20. So the probability that the segments overlap is $2/9$.